ECE-118/218 Introduction to Mechatronics

Uno32 Stack: IO Board and Power Distribution

University of California Santa Cruz — Spring 2026

Overview

The ChipKit Uno32 module is a single-board computer produced by Digilent, Inc., which integrates a Microchip PIC3 microcontroller (PIC32MX320F128H), FTDI USB-to-serial converter, voltage regulation, and 3.3V clamp diodes into a compact package. ECE118 uses the Uno32 platform throughout the lab exercises and final project. The Uno32 is an Arduino-compatible module, though we reflash the Arduino firmware to use a custom bootloader that allows us to program the PIC32 directly in C over the USB link.

However, the Uno32 was not designed with ECE118 in mind, so it is not exactly what we want. To get closer to the ideal platform for the course, we designed and manufactured the Uno32 I/O board.

The Uno32 I/O Board has several major functions:

- Bring out all of the connections from the Uno32 to clearly defined and easily accessible ports with uniform connectors.
- Provide a wide range of voltage input through an onboard switching regulator (6V to 30V).
- Provide over-voltage and over-current protection circuitry on all port pins.
- Provide LED banks for user visual outputs.
- Provide both local and remote on/off power and reset switches.
- Provide reverse polarity protection.
- Provide fuse-protected power distribution for additional circuits.

The data sheet for the ChipKit Uno32 can be found on the Digilent website at: `https://digilent.com/reference/chipkit_uno32/refmanual`

Bumps and Road Hazards The Uno32 Stack should always have the miniUSB cable threaded inside the standoffs (this provides strain relief); do not disassemble it. The IO sheild clamps will **ONLY** work if the board is powered through the Power Distribution Board—if you are powering it through the USB, then it is unprotected. Lastly, you need the SPI jumpers in the IO board or you will get strange behavior.

Using the ECE118 Uno32 I/O Board

Using the ECE118 Uno32 I/O Board should be straightforward. All the connections that will typically be made to the board are located around the edge of the board, and are clearly labeled and logically grouped. At the end of this document is a table that lists each pin on the ports, along with its associated pins on the PIC32 microcontroller. All ports are clamped with diodes (TL7226) and series resistors to protect them from overvoltage, undervoltage, or excessive current.

Digital Input/Output Ports (`PORT-X`, `PORT-Y`, and `PORT-Z`)

Digital input and output ports are located on the three ports on the top of the board. They can be accessed either directly at the bit level through the `#defines` in the `PORTS` library, or as word-sized ports through the functions contained within the library. Following the convention that Microchip uses, the `_TRIS` registers are used to set output (`0`) or input (`1`) for each pin. The `_LAT` registers are used to write the output, and the `_BIT` registers to read from the port.

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Port X, Y, and Z each have the outer two pins of the connectors grounded at the I/O board.

The pinout of PORT-X, PORT-Y, and PORT-Z is as follows:

PORT	Connection
Pin 1	GND
Pin 2	GND
Pin 3–12	PORT-[X,Y,Z] 3–12
Pin 13	GND
Pin 14	GND

Digital and Analog Input/Output Ports (PORT-V and PORT-W)

Analog inputs (10-bit ADC) are available on PORT-V and PORT-W, and can be set either as analog or digital I/O. The analog pins can be accessed through the AD library, and the digital I/O can be accessed through the PORTS library.

Port V and W each have the outer pins connected to ground, and only have six active pins each, with the remaining pins connected to ground.

The pinout of PORT-V and PORT-W is as follows:

PORT	Connection
Pin 1	GND
Pin 2	GND
Pin 3–8	PORT-[V,W] 3–8
Pin 9	GND
Pin 10	GND
Pin 11	GND
Pin 12	GND
Pin 13	GND
Pin 14	GND

Battery Voltage Monitoring

The Uno32 I/O Board includes a precision 10:1 voltage divider connected to an ADC pin that is accessible through the AD library.

LED Banks

The Uno32 I/O Board includes 12 LEDs organized into 3 banks of 4 LEDs each. They are connected to a subset of pins (see the connection table at the end of this document) and are accessible through the LED library. The LED CONTROL jumper (JP7) connects the top of each bank of LEDs to the 3.3V rail through a set of current-limiting resistor banks. Leave the jumper open if you want to use those pins as normal I/O pins.

Reset and Remote Reset (SW1 and JP3)

The Uno32 I/O Board includes a reset switch (SW1) as well as a remote reset jumper (JP3). The switch is a momentary-contact switch which holds the PIC32 in reset while depressed. The remote switch works by shorting either of the outer pins to the center pin to reset the microcontroller. The microcontroller will be held in reset as long as the pins are shorted.

Power and Remote Power (SW1 and JP1 on lower board)

The Uno32 I/O Board includes a power switch (SW1 on the lower board) as well as a remote power jumper (JP1) with three right-angle pins. The switch is a side-slider switch, labeled ON and OFF, which connects the power line back to the source. The remote switch works by shorting either of the outer pins to the center pin to power the board.

SPI Master/Slave Select Jumper (J2)

The Uno32 I/O Board moves the jumpers used to select SPI master or slave from the ChipKit board to the I/O board with exactly the same functionality. Both jumpers should be moved together and set toward the top of the board for SPI master (normal), or both down for SPI slave. Note that the PORTS and LED libraries both have an SPI_MASTER #define that should be commented out for SPI slave operation.

Power Regulation and Protection Circuitry

The Uno32 I/O Board is powered through connector J1 on the bottom power distribution board. The connector is reverse-polarity protected via a dual P-channel FET that connects the power line when connected correctly.

Uno32 I/O Board Power Supply Requirements: $+6\text{V} < V_{in}$ **(J1, pin 1)** $< +30\text{V}$

The input voltage is regulated via a TI TPS5410 switching regulator, which provides clean 5V to the clamps and the ChipKit board. The regulated 3.3V power rail for the microcontroller comes from the LDO on the ChipKit board itself. The I/O board has its power smoothed via two inductors, one on the power line and one on the ground line. The power distribution board is fused on each half with 3A fast-blow 2AG fuses.

There are *power good* LEDS that indicate that input power, and both fuses are intact. If you blow a fuse, SLOW DOWN! and figure out why it blew rather than simply plugging the circuit into the other side.

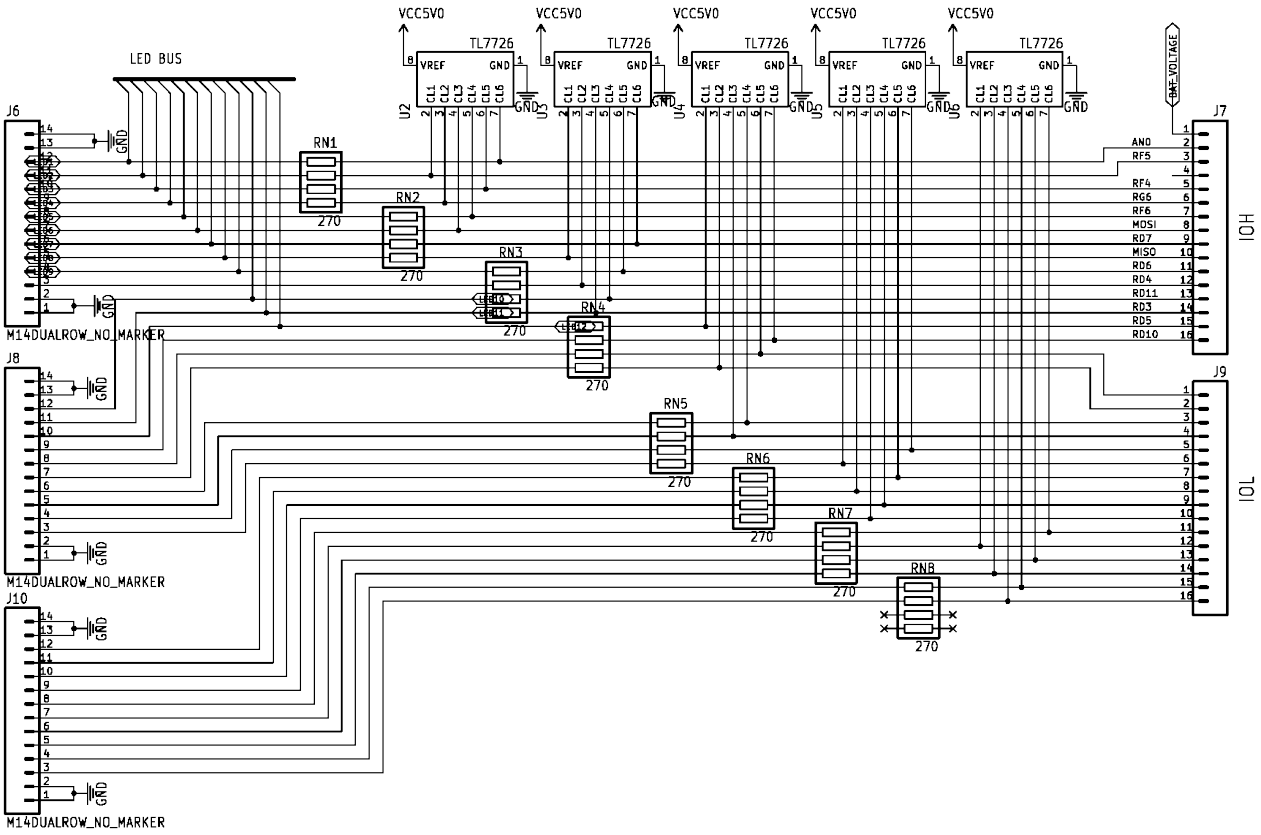
J1	Connection
Pin 1	Power (6V to 30V)
Pin 2	High Current Ground

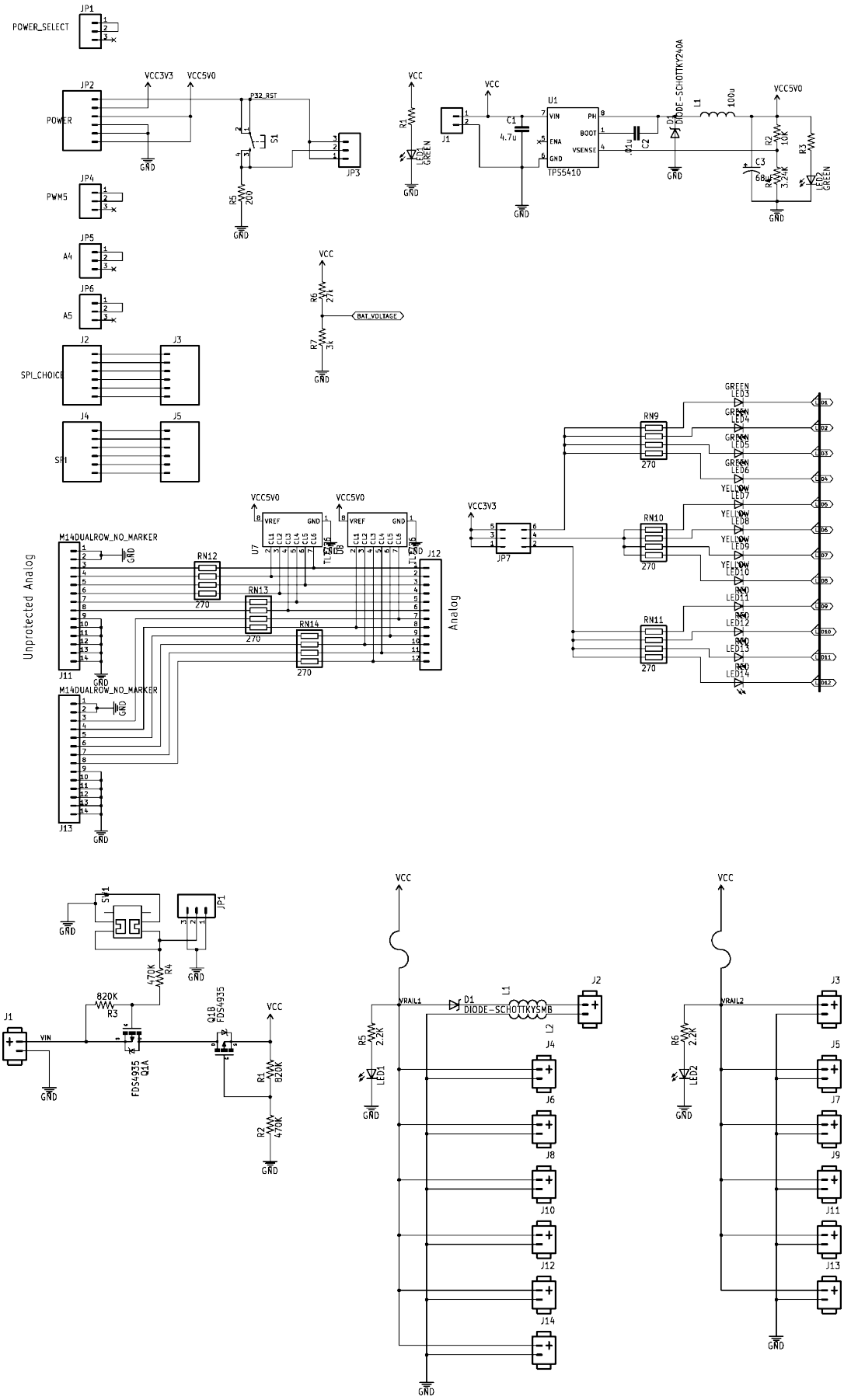
ECE118 Uno32 I/O Board Connection Table

PIC32 Pin #	Connector Pin #	chipKIT Pin #	PIC32 Signal	Protection Pin #	Notes
15	J5-16	41	PGC1/AN1/VREF-/CVREF-/CN3/RB1	Battery Voltage	Inaccesible from I/O
14	J7-01	A00/14	C2IN-/AN2/SS1/CN4/RB2	PORT_V-3	RC Servo
13	J7-02	A06/20	C2IN+/AN3/CN5/RB3	PORT_V-4	RC Servo
12	J7-03	A01/15	C1IN-/AN4/CN6/RB4	PORT_V-5	
11	J7-04	A07/21	C1IN+/AN5/CN7/RB5	PORT_V-6	
21	J7-05	A02/16	U2CTS/C1OUT/AN8/RB8	PORT_V-7	
22	J7-06	A08/22	PMA7/C2OUT/AN9/RB9	PORT_V-8	
24	J7-08	A09/23	TDO/PMA12/AN11/RB11	PORT_W-3	
23	J7-07	A03/17	TMS/CVREFOUT/PMA13/AN10/RB10	PORT_W-4	
28	J7-10	A10/24	TDI/PMA10/AN13/RB13	PORT_W-5	
27	J7-09	A04/18	TCK/PMA11/AN12/RB12	PORT_W-6	
30	J7-12	A11/25	PMALL/PMA0/AN15/OCFB/CN12/RB15	PORT_W-7	RC Servo
29	J7-11	A05/19	PMALH/PMA1/U2RTS/AN14/RB14	PORT_W-8	RC Servo
32	J5-14	40	PMA8/U2TX/SCL2/CN18/RF5	PORT_X-03	LED_BANK3_2, RC Servo
16	J5-15	42	PGED1/PMA6/AN0/VREF+/CVREF+/CN2/RB0	PORT_X-04	LED_BANK3_3, RC Servo
4	J5-11	13	SCK2/PMA5/CN8/RG6	PORT_X-05	LED_BANK3_0
31	J5-12	39	PMA9/U2RX/SDA2/CN17/RF4	PORT_X-06	LED_BANK3_1
5	J5-09 or J5-07	12	SDI2/PMA5/CN8/RG7	PORT_X-07/9	LED_BANK2_2*
35	J5-10	38	U1RTS/BCLK1/SCK1/INT0/RF6	PORT_X-08	LED_BANK2_3
6	J5-07 or J5-09	11	SDO2/PMA3/CN10/RG8	PORT_X-09/7	LED_BANK2_0*
55	J5-08	37	CN16/RD7	PORT_X-10	LED_BANK2_1
52	J5-05	10	PMWR/OC5/IC5/CN13/RD4	PORT_X-11	PWM5

PIC32 Pin #	Connector Pin #	chipKIT Pin #	PIC32 Signal	Protection Pin #	Notes
54	J5-06	36	CN15/RD6	PORT_X-12	LED_BANK1_3
45	J5-04	35	IC4/PMCS1/PMA14/INT4/RD11	PORT_Y-03	LED_BANK1_2
51	J5-03	9	OC4/RD3	PORT_Y-04	PWM4, LED_BANK1_1
53	J5-02	34	PMRD/CN14/RD5	PORT_Y-05	LED_BANK1_0
44	J5-01	8	IC3/PMCS2/PMA15/INT3/RD10	PORT_Y-06	RC Servo
3	J6-16	33	PMD7/RE7	PORT_Y-07	RC Servo
43	J6-15	7	IC2/U1CTS/INT2/RD9	PORT_Y-08	
2	J6-14	32	PMD6/RE6	PORT_Y-09	
50	J6-13	6	OC3/RD2	PORT_Y-10	PWM3
1	J6-12	31	PMD5/RE5	PORT_Y-11	
49	J6-11	5	OC2/RD1	PORT_Y-12	PWM2
64	J6-10	30	PMD4/RE4	PORT_Z-03	
59	J6-09	4	RF1	PORT_Z-04	
63	J6-08	29	PMD3/RE3	PORT_Z-05	
46	J6-07	3	OC1/RD0	PORT_Z-06	PWM1
62	J6-06	28	PMD2/RE2	PORT_Z-07	
42	J6-05	2	IC1/RTCC/INT1/RD8	PORT_Z-08	RC Servo
61	J6-04	27	PMD1/RE1	PORT_Z-09	RC Servo
33	J6-03	1	U1TX/SDO1/RF3	PORT_Z-10	Serial (unusable)
60	J6-02	26	PMD0/RE0	PORT_Z-11	
34	J6-01	0	U1RX/SDI1/RF2	PORT_Z-12	Serial (unusable)

ECE118 Uno32 I/O Board and Power Distribution Board Schematics




POWER_SELECT
JP1
POWER
JP2
VCC3V3
VCC5V0
P32_RST
S1
JP3
PWM5
JP4
R5
200
GND
A4
JP5
A5
JP6
J2
J3
SPI_CHOICE
J4
J5
SPI
VCC
R6
27K
BAT_VOLTAGE
R7
3K
R1
LED
GREEN
J1
U1
VIN
PH
BOOT
ENA
VSENSE
GND
TPS5410
C1
4.7u
.01u
C2
D1
DIODE-SCHOTTKY2A0A
L1
100u
C3
68u
R4
3.24K
R2
10K
R3
LED
GREEN
JP7
VCC3V3
RN9
RN10
RN11
270
GREEN
LED3
LED4
LED5
LED6
YELLOW
LED7
LED8
LED9
LED10
360
LED11
LED12
LED13
LED14
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
Unprotected Analog
M14DUALROW_NO_MARKER
J11
J13
RN12
RN13
RN14
U7
VREF
GND
TL431
D5
J12
Analog
SW1
JP1
470K
R4
820K
R3
Q1A
FDS4935
Q1B
FDS4935
R1
820K
R2
470K
VIN
VRAIL1
VRAIL2
D1
DIODE-SCHOTTKYSMB
L1
L2
R5
2.2K
R6
2.2K
LED1
LED2
J2
J3
J4
J5
J6
J7
J8
J9
J10
J11
J12
J13
J14